USN. | | | | | | | | | | [21JUOE1001/22JUOE1001]

THIRD SEMESTER M.Tech EXAMINATION, DECEMBER 2024

BUSINESS ANALYTICS

Time: 3 Hours **Maximum Marks: 100**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a**	Explain **i.** Nominal data **ii.** Interval data	**L2**	**CO1**	**05**
	b	Discuss the use of prescriptive analytics in business analytics.	**L2**	**CO1**	**05**
	c	Describe box plot and its applications.	**L2**	**CO1**	**07**
	d	Discuss the data organizing method in a relational database.	**L2**	**CO1**	**03**
2.	**a**	Explain snowflake schema used in data modeling.	**L2**	**CO2**	**10**
	b	Describe the different measure of location used to analyze data.	**L2**	**CO2**	**05**
	d	Explain the cluster sampling method.	**L2**	**CO2**	**05**
3.	**a**	Explain the binomial distribution and its applications.	**L2**	**CO3**	**06**
	b	What is a confidence interval? How is it useful?	**L1**	**CO3**	**06**
	d	Describe the Hypothesis-Testing procedure.	**L2**	**CO3**	**08**
		OR			
4.	**a**	Explain the weighted moving average method for time series prediction.	**L2**	**CO3**	**08**
	b	Differentiate between cross-sectional and time series data.	**L2**	**CO3**	**08**
	c	Describe the mathematical principle behind linear regression.	**L2**	**CO3**	**04**
5.	**a**	Explain the difference between qualitative and quantitative forecasting giving an example for each.	**L2**	**CO4**	**08**
	b	Explain seasonal time series pattern and its application.	**L2**	**CO4**	**08**
	c	Describe a polynomial trend line.	**L2**	**CO4**	**04**
		OR			
6.	**a**	Describe the steps in building an optimization model.	**L2**	**CO4**	**05**
	b	Discuss the decision variable, objective function and constraints for the following models **i.** Product mix. **ii.** Blending. **iii.** Transportation.	**L2**	**CO4**	**10**

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	c	Explain Binding and non-binding constraints.	L2	CO4	05
7.	a	Explain the payoff in decision analysis.	L2	CO5	06
	b	Explain the minimin strategy.	L2	CO5	06
	c	Describe opportunity loss.	L2	CO5	04
	d	What is a branch in a decision tree?	L1	CO5	04
		OR			
8.	a	Explain the following in context of data mining **i.** Association. **ii.** Cause effect modelling.	L2	CO5	10
	b	Describe the objective of cluster analysis.	L2	CO5	05
	c	What are the applications of data mining?	L2	CO5	05